

EPPS 7313 Final Report

Hypothesis:

H0: Study hours do not affect exam scores even when accounting for hours of sleep.

H1: Study hours affect exam scores even when accounting for hours of sleep.

Variables:

Dependent Variable: Exam Performance

Independent Variable: Study Time

Control Variable: Sleep Time

Dataset:

The dataset I used was provided by the web platform Kaggle and it is a randomly generated data that tracks student performance factors. The dataset had 6607 total entries and 20 columns including: Hours Studied, Attendance Rate, Parental Involvement, Access to Resources, Extracurricular Activities, Sleep Hours, Previous Exam Scores, Motivation Level, Internet Access, Tutoring Sessions, Family Income, Teacher Quality, School Type, Peer Influence, Physical Activity, Learning Disabilities, Parental Education Level, Distance from Home, Gender, and Current Exam Scores. The variables necessary for this analysis were the dependent variable: Current Exam Scores, the independent variable: Hours Studied, and the control variable: Sleep Hours.

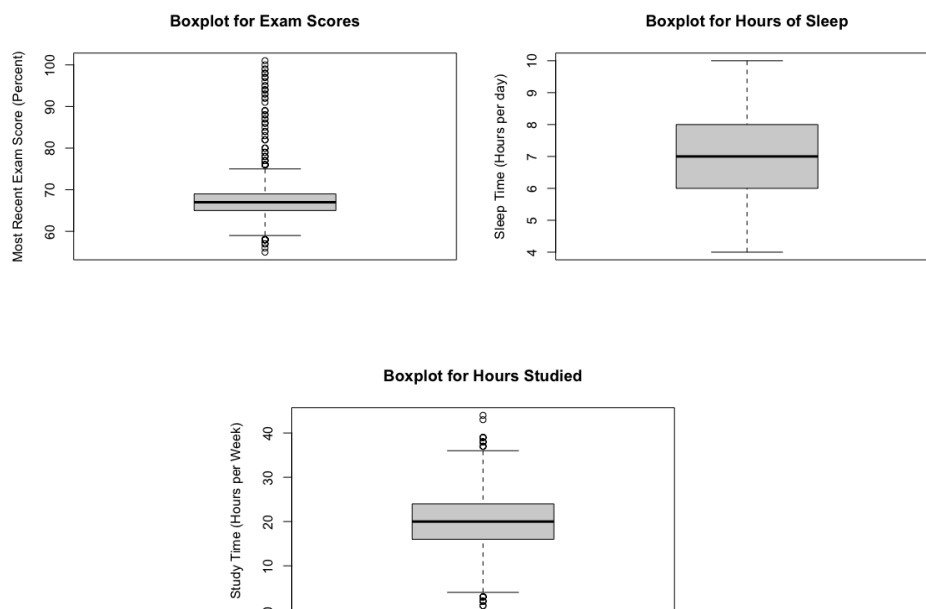
Statistical Analysis:

To begin, the dataset was uploaded into R and summary statistics were obtained for each variable.

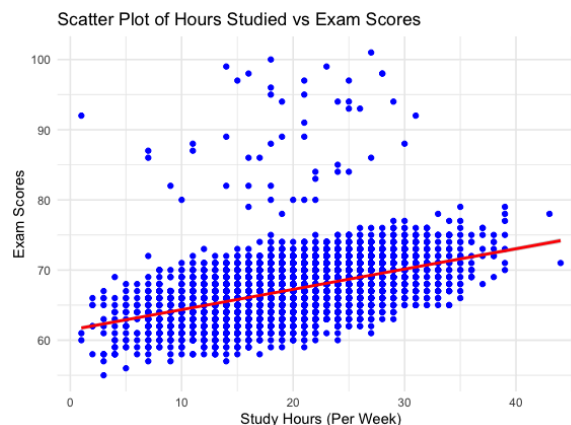
Hours_Studied					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.00	16.00	20.00	19.98	24.00	44.00
Exam_Score					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.

55.00	65.00	67.00	67.24	69.00	101.00
Sleep_Hours					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
4.000	6.000	7.000	7.029	8.000	10.000

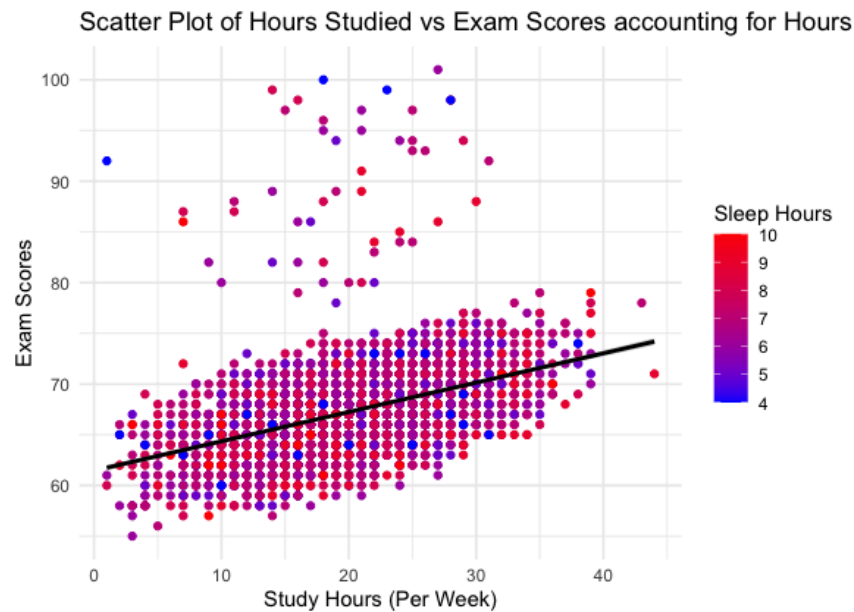
Then I visualized the summary statistics using boxplots for all three variables.



The data was randomly generated however some concerns for this data can include selection maturity as the participants would be children who are naturally prone to regressing to the mean or maturing as they age. The boxplot for Hours of Sleep shows signs of normality but the Exam Scores are less so. By creating a scatterplot, I was able to see the relationship between Hours Studied and Exam Scores. As you can see there was a positive linear relationship between the variables which was further proved through correlation test which showed a p-value of $2.2e-16$. The Pearson Correlation Coefficient was 0.44544 which shows moderate positive correlation.



After looking at the dependent and independent variables, I did a second scatterplot attempting to visualize data while controlling for Hours of Sleep.



Significance tests done while including the control variable show that study hours when accounting for sleep is significant at the .05 level with a value of $2e-16$. The P-value for Sleep hours is .0467 which is also significant at the .05 level and implies that hours of sleep has a significant impact on exam scores.

Finally, I did regression analysis and the R-squared was 0.1983 which means that 19.83% of variations within the exam scores were explained by the model.

```
Residuals:
    Min       1Q   Median       3Q      Max
-8.532 -2.243 -0.111  2.046 33.493
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    61.456984   0.149196   411.92  <2e-16 ***
sleep_data$Hours_Studied  0.289291   0.007154   40.44  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.483 on 6605 degrees of freedom
Multiple R-squared:  0.1984,    Adjusted R-squared:  0.1983
F-statistic: 1635 on 1 and 6605 DF,  p-value: < 2.2e-16
```

R Code:

```
sleep_data <- read.csv("sleep.csv")
View(sleep_data)
summary(sleep_data)
summary(sleep_data$Hours_Studied)
summary(sleep_data$Exam_Score)
summary(sleep_data$Sleep_Hours)
boxplot(sleep_data$Hours_Studied,
        main = "Boxplot for Hours Studied",
        ylab = "Study Time (Hours per Week)")
boxplot(sleep_data$Exam_Score,
        main = "Boxplot for Exam Scores",
        ylab = "Most Recent Exam Score (Percent)")
boxplot(sleep_data$Sleep_Hours,
        main = "Boxplot for Hours of Sleep",
        ylab = "Sleep Time (Hours per day)")
### Data Visualization
install.packages("ggplot2")
library(ggplot2)
ggplot(sleep_data, aes(x = Hours_Studied, y = Exam_Score)) + # Set the variables
  geom_point(color = "blue", size = 1.5) + # Set the color and size of points # Set
the location of texts
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  labs(
    title = "Scatter Plot of Hours Studied vs Exam Scores", # Set the title of the
plot
    x = "Study Hours (Per Week)", # Set the title of x-axis
    y = "Exam Scores", # Set the title of y-axis
  ) +
  theme_minimal() # Apply a minimal and clean style
### Significance test
cor.test(sleep_data$Hours_Studied, sleep_data$Exam_Score)
# p-value: 2.2e-16 < 0.05, reject the null hypothesis
# There is statistically significant relationship
# Correlation Coefficient 0.445455 > 0, moderate positive correlation
### Data Visualization with Control Variable
ggplot(sleep_data, aes(x = Hours_Studied, y = Exam_Score, color = Sleep_Hours)) + #
Set the variables
  geom_point(size = 1.5) + # Set the color and size of points # Set the location of
texts
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  scale_color_gradient(low = "blue", high = "red") +
  labs(
    title = "Scatter Plot of Hours Studied vs Exam Scores accounting for Hours of
Sleep", # Set the title of the plot
    x = "Study Hours (Per Week)", # Set the title of x-axis
    y = "Exam Scores", # Set the title of y-axis
    color = "Sleep Hours"
  ) +
  theme_minimal() # Apply a minimal and clean style
### Significance test with control variable
lm_control2 <- lm(Exam_Score ~ Hours_Studied + Sleep_Hours, data = sleep_data)
summary(lm_control2)
# P-value for study hours when accounting for sleep is significant as 2e-16<.05.
# P-value for Sleep hours is 0.0467 which means it has a significant impact on exam
scores as well.
### Regression Analysis
regressionmodel <- lm(sleep_data$Exam_Score ~ sleep_data$Hours_Studied)
summary(regressionmodel)
```

References:

Kaggle. (2024). *Student Performance Factors*. Retrieved December 13, 2024, from <https://www.kaggle.com/datasets/lainguyn123/student-performance-factors/data>